# Your Title: What You Did on DermaMNIST

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TABLE I
BALANCED ACCURACY (MEAN ± STD OVER 3 SEEDS)

| variant | val | test | 95 % interval vs. A (val) |
|---|---|---|---|
| A: baseline | 0.46 ± 0.02 | – | – |
| B1: class weights | 0.53 ± 0.01 | – | [0.02, 0.13] |
| B2: ... | | | |

***Abstract*—3–5 sentences: the task, your baseline, your improvements, your main result on the test set (balanced accuracy, mean ± std over seeds), and the one thing you learned.**

## I. INTRODUCTION

Why is this task relevant? What is hard about it (7 classes, strong imbalance)? What did you try, and what is your main finding? (about 1/3 column)

## II. DATA

DermaMNIST [1]: number of images per split, the 7 classes, the class counts (a small table or bar plot), example images. How did you preprocess the images (normalisation, augmentation)?

## III. METHODS

### A. Baseline (A)

The CNN of the starter notebook: layers, number of parameters, loss, optimiser, learning rate, epochs.

### B. Improvements (B)

Describe every change and *why* you made it. At least one must be motivated by a paper or another good source — cite it [2].

### C. Evaluation

Main metric: balanced accuracy. Also macro-F1 and the sensitivity per class. Models and settings were chosen on the validation set; every variant was trained with at least 3 seeds. Differences to the baseline: 95 % bootstrap interval of the difference on the validation set.

## IV. RESULTS

Table I shows mean ± std over seeds. Change one thing at a time.

Add one key figure, e.g. the confusion matrix of your final model or the learning curves:

## V. DISCUSSION

What worked, what did not, and why? Which classes are still hard? What would you try next? Limitations (small images, one dataset, ...).

## VI. CONCLUSION

2–3 sentences.

## AI USE STATEMENT

Which AI tools did you use, and for what (e.g. "ChatGPT to explain an error message", "Copilot for plotting code")? If none: "No AI tools were used."

## CONTRIBUTION STATEMENT

Who did what (2–3 sentences), e.g. "A. wrote the data pipeline and the augmentation experiments; B. implemented the class weights and the evaluation; both wrote the report and the poster."

## REFERENCES

[1] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2 - a large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, no. 1, p. 41, 2023.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.